

TROY WU

troywu100@gmail.com | +1 (236) 591-5923 | linkedin.com/in/troy-wu | github.com/troy-wu | troy-wu.github.io

EDUCATION

University of British Columbia

Expected Graduation: May 2027

Bachelor of Computer Science and Business | GPA: 3.85/4.00

Vancouver, BC

- Dean's Honour Roll, Trek Excellence Scholarship (Top 5%), Ranked 6/791 in UBC Sauder.
- Courses:** Data Structures & Algorithms; OOP (Java); Operating Systems; Relational Databases.

Tsinghua University

February 2025 - June 2025

School of Economics and Management | Exchange Term

Beijing, China

EXPERIENCE

Amazon

May 2026 - August 2026

Incoming Software Development Engineer Intern

Vancouver, BC

D2L (Desire2Learn)

January 2026 - April 2026

Software Engineer Intern - Lumi (Brightspace GenAI)

Vancouver, BC

- Developed full-stack GenAI features for LMS workflows deployed across **400+ institutions** and **260K+ annual generations** using Web Components/LitElement, C# MVC, and SQL databases.
- Built Brightspace's **AI Audio Overview** feature end-to-end, delivering natural AI-narrated content summaries via serverless **Claude Sonnet + AWS Bedrock/Lambda and Polly TTS** pipeline to support diverse learning styles.
- Refined and evaluated prompt strategies for generative AI features in D2L Brightspace, **reducing malformed outputs by 30%** and improving content consistency across subjects.
- Built and refactored Playwright E2E tests for critical GenAI workflows **expanding test coverage by 28%**, and consolidated duplicate code through shared TestBench fixtures **reducing suite runtime by 32%**.

Absolute Security

May 2024 - December 2024

Software Engineer Intern

Vancouver, BC

- Built and deployed full-stack, telemetry security services (Java, JavaScript/Manifest V3 extension) for **1M+** enterprise Chromebook endpoint devices, **enabling continuous monitoring and threat detection**.
- Engineered a scalable Java microservice integrated with **AWS Grafana** to automate and standardize log exports for **1M+** devices, **boosting error detection speed by 30%** and QA efficiency.
- Rebuilt QA log interface (HTML, CSS, JS), **reducing export latency by 50%** and improving usability.
- Developed robust JUnit/Mocha test suites (96.8% coverage) ensuring reliability of production services.
- Applied agile development practices using Jira, and contributed to deployment pipelines utilizing Argo, Docker, and Kubernetes to support scalable enterprise solutions.

PROJECTS

BasketBot – ProduHacks 2024 Winner 🏆 | Python, OpenCV, TensorFlow, PyAudio, VOSK

- Winner of ProduHacks (50 teams, \$400) by building an object-detection-based automatic basketball scorekeeper.
- Implemented TensorFlow-based machine learning models to recognize basketball nets for accurate scoring.
- Integrated OpenCV computer vision with speech recognition using PyAudio and the VOSK library.

Campus Explorer 🏠 | TypeScript, Node.js, Express.js, JavaScript, HTML/CSS, Google Maps API

- Built full-stack web application with TypeScript/Express.js backend and interactive JavaScript frontend enabling real-time querying and visualization of 300+ campus rooms via Google Maps API.

BattleIQ – Clash Royale Insights Dashboard 📊 | Python, REST API, Data Analysis

- Engineered a Python data processing system that aggregated and analyze Clash Royale battles via REST API.

LEADERSHIP

BUCS Community Executive | UBC Business and Computer Science Major Community Team

- Increased YoY engagement by 31% for 500+ students in BUCS by planning & organizing community events.

TECHNICAL SKILLS

Languages: Python, Java, C#, TypeScript, JavaScript, C++, SQL.

Frameworks/Libraries: Chai, Mocha, React.js, Node.js, Express.js, Sinon, OpenCV, Flask, JUnit.

Developer Tools: Docker, Kubernetes, ArgoCD, Databricks, Git, Gitlab, MongoDB, Jira, Github, OracleDB.